

Ch-2 : Regular languages & Finite Automata (25% weightage)

Conversion Questions

- ① Strings/Language $\rightarrow$ RE (4 Marks)
 - ② Strings/Language $\rightarrow$ FA (3, 4, 7 Marks)
 - ③ RE $\rightarrow$ FA (7 Marks)
 - ④ RE $\rightarrow$ NFA- λ (3, 4, 7 Marks)
 - ⑤ NFA- $\lambda \rightarrow$ NFA (7 Marks)
 - ⑥ NFA $\rightarrow$ FA/DFA (7 Marks)
 - ⑦ NFA- $\lambda \rightarrow$ FA/DFA (7 Marks)
 - ⑧ DFA $\rightarrow$ Minimization of DFA (7 Marks)
 - ⑨ Moore $\rightarrow$ Mealy
 - ⑩ Mealy $\rightarrow$ Moore
- } (3, 4, 7 Marks)

Note:

FA = DFA to solve Qs in Exam.

Other Questions :

- ⑪ Draw FAs of Union, Intersection & difference.
(e.g. $L_1 \cup L_2$, $L_2 \cap L_1$, $L_1 - L_2$, etc.) (7 Marks)
- ⑫ Find ϵ -closure of NFA- λ (3 Marks)
- ⑬ Pumping Lemma (3, 7 Marks)
- ⑭ Explain that given language is not regular. (3 Marks)
- ⑮ Kleene's theorem (7 Marks)
- ⑯ Definitions of FA, DFA, NFA, NFA- λ , ϵ -closure, (3 Marks)
Regular language, Regular Expression with example.
- ⑰ Difference between NFA & DFA (3 Marks)
Difference between Moore & Mealy Machine (3 Marks)
- ⑱ Applications of FA & REs. (3 Marks)